

A Dormant Subspace, Not a Missing One: A Decodable but Causally Inert Answer-Site Representation of Arithmetic in Llama-3.1-8B

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Abstract

Large language models often fail at arithmetic composition even when they solve every part in isolation. We study a token-matched instance in Llama-3.1-8B: wrapping a multiplicative operand in an additive identity, $(0 + B) * C$, collapses multiplication accuracy (instruct 0.27, base 0.51) although the model evaluates the wrapped sub-expression perfectly (1.00) and multiplies inside a bracket (0.91). The failure is operation-specific (the additive analogue composes at 0.99), depth-sensitive, magnitude-controlled, and recovers with a few in-context examples. We then ask what the internal representation looks like. At the answer position the operand B and the near-product $B \cdot C$ are linearly decodable ($R^2 \approx 0.96$), which invites a “the value is there but unused” story. We show this reading is an interpretability illusion. The decode is operand-dominated: it has no margin over a linear- (B, C) baseline where there is no headroom, and only a small margin (gap ≈ 0.06 , CI excludes zero) where there is. More decisively, the decodable direction is causally inert: steering along it does not change the output, and a full-residual swap at the answer position does not flip it either, while patching the operand positions does move it (recovery 0.50), a dead site, not a dead probe. Decodability at the answer site is not evidence of computation there. The collapse replicates across model families in matched format, with a prompt-format confound we make explicit.

1 Introduction

Language models solve many arithmetic problems yet fail on superficially trivial variants. A recurring explanation is that some intermediate value is computed but not propagated, tested by probing the residual stream: if the value decodes, the model “knows” it. We show this inference is unsafe. A decodable value at the answer position can be an operand-reconstructible, causally inert shadow of the computation, not the computation itself.

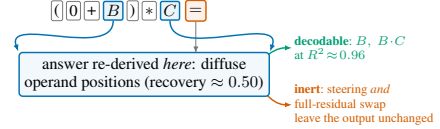


Figure 1: The answer-site subspace is *dormant*, not missing. At the “=” position the operand and near-product are linearly decodable, but that direction is causally inert (steering and a full-residual swap change nothing) and operand-explained; the answer is instead re-derived diffusely from the operand positions. Decodability at the answer site is not evidence of computation there.

Our vehicle is a single controlled failure. For $B, C \in [20, 49]$, wrapping the first operand in an additive identity, $(0 + B) * C =$, collapses accuracy to 0.27 (instruct) / 0.51 (base) although the model returns B for $(0 + B) =$ at 1.00 and multiplies inside a bracket, $(B * C) =$, at 0.91; the gap is large and tight ($\Delta(C4 - C1) = 0.64$, 95% CI $[0.60, 0.69]$, instruct). This is a clean hook, not our contribution: we use it to ask whether a decodable representation at the answer site licenses a claim about computation. Both B and the product $B \cdot C$ are linearly decodable there ($R^2 \approx 0.96$), inviting the reading that the product “is represented but not used.” Figure 1 summarizes why that reading fails.

Contributions.

- A token-matched, control-rich instance of a compositional arithmetic blind spot in Llama-3.1-8B: operation-specific, depth-sensitive, magnitude-controlled, and few-shot-recoverable, each with a confidence interval (§4).
- Evidence that the decodable answer-site subspace is *operand-dominated*: it has no margin over a linear- (B, C) baseline where the band leaves no headroom, and only a small margin (gap ≈ 0.06 , CI excludes zero) where it does, robust to a pre-registered probe layer (§5).

- A causal test showing the decodable direction is *inert*: probe-direction steering and a full-residual swap at the answer position both leave the output unchanged, anchored by a positive control (the same-family patching moves the answer at the operand positions, recovery 0.50), so the null is a dead site, not a dead probe. Within the swap’s small sub-flip effect, $\approx 85\%$ of the variance is orthogonal to the decode direction (§5).
- A cross-family replication in matched bare format, together with an explicit prompt-format confound that we argue precludes reading the cross-model results as a clean family dissociation (§6).

We read the answer-site subspace as *dormant* in the sense of Makelov et al. (2024): present to a probe, invisible to the computation.

2 Related Work

Surface form and mathematical brittleness.

Model math accuracy is sensitive to surface structure that leaves the answer invariant. Williamson et al. (2025), at the same venue, characterize a behavioral blind spot in natural-language word problems where failure tracks dependency-locality complexity and recovers under structure-lowering rephrasing, and conjecture a coupling between surface form and internal representation. Our instance is orthogonal on both axes and *tests* that conjecture: a token-matched *symbolic* expression whose additive-identity wrapper carries no dependency-locality cost, for which we show the obvious representational reading (a computed-but-unused product at the answer site) does not hold; the internal signature is a dormant, operand-explained subspace, not a missing representation. Reusch and Lehner (2022) show operator-tree structure is decodable from embeddings; we show that answer-site decodability of the arithmetic *value* can be an operand-reconstructible artifact that is causally inert, so decodability of arithmetic structure does not certify its use.

Decodability versus causal use. A linear probe measures correlation, not use (Belinkov, 2022), and can succeed on random labels without a control (Hewitt and Liang, 2019). The sharpest form is the subspace-patching illusion of Makelov et al. (2024): a direction can carry a decodable signal yet be causally disconnected from the output. Our

result is an arithmetic instance of this illusion, established with a baseline and a steering test rather than assumed. Matsumoto et al. (2022) find decoded intermediate values in neural math solvers to be causally usable; we find a decodable answer-site subspace that is causally inert. The results are compatible: whether a decodable value is used is a per-site empirical question.

Mechanisms of arithmetic. Stolfo et al. (2023) trace arithmetic from operand tokens to the final position through late modules, and Nikankin et al. (2025) argue models use a diffuse “bag of heuristics.” Our localization agrees, and adds that the decodable answer-site value, easily mistaken for the computed result, is a byproduct of that operand information. For method we follow patching best practice (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024): symmetric counterfactuals, a full-answer log-probability metric, and a positive control per null.

3 Setup and Research Questions

Model and stimuli. We study Llama-3.1-8B (Grattafiori et al., 2024), base and instruction-tuned, in bare continuation format, by greedy decoding and integer parsing. Stimuli are $N = 400$ operand pairs (B, C) drawn once with a fixed seed from $[20, 49]$ and held constant across every condition, token-length matched, with spacing controlled. The central contrast is $C1 = (0 + B) * C =$ (the composed failure) against $C4 = (B * C) =$ (multiply inside a bracket, succeeds) and $C6 = (0 + B) =$ (evaluate the wrapper, succeeds).

Metrics. Accuracies carry Wilson intervals; paired contrasts carry paired bootstrap intervals (10,000 resamples). Decodability is five-fold cross-validated ridge R^2 (item-bootstrap interval) at a named token position. Selectivity compares the decode against a linear- (B, C) baseline (a probe given only the raw operands and Gaussian noise), so a high R^2 that is merely linear in the operands does not count as representing the product. Causal tests use symmetric operand counterfactuals and a full-answer log-probability metric $\Delta = \log p(P') - \log p(\text{true})$; because Llama chunks the leading digits, a first-token score is non-discriminative between two products.

Research questions. **RQ1:** is the composition failure a genuine, controlled effect or an artifact of difficulty, magnitude, or elicitation (§4)? **RQ2:**

	surface	base	instruct
C0	$B * C =$	0.838	0.848
C4	$(B * C) =$	0.890	0.908
C6	$(0 + B) =$	1.000	1.000
C1	$(0 + B) * C =$	0.508	0.265
A1	$(0 + B) + C =$	n/a	0.995
D1	$((0 + B)) * C =$	n/a	0.038

Table 1: The composition blind spot (exact-match accuracy, $N=400$). The parts succeed (C4, C6) but the composition collapses (C1); $\Delta(C4-C1) = 0.64$, CI [.60, .69] (instruct). A1/D1 are instruct-only. Full battery with Wilson intervals in Appendix A.

is the near-product decodable at the answer site a *represented* quantity or reconstructible from the operands alone (§5)? **RQ3**: is the decodable direction causally used to produce the answer (§5)?

4 The Blind Spot

The model evaluates every ingredient of $(0 + B) * C$ yet fails to compose them (Table 1). We establish that this is a real effect, not an artifact, with four controls, and we state each at the precision the data support.

It is operation-specific, not format brittleness. Replacing the outer multiplication with addition, $(0 + B) + C =$, restores accuracy to 0.995 (instruct) against 0.265 under *identical* format; $\Delta(A1 - C1) = 0.730$, 95% CI [0.69, 0.77]. It is also spacing-invariant: with all spaces removed the inside-bracket multiply still survives (C8 = 0.785 instruct, 0.765 base) while the composition still collapses (C7 = 0.19, 0.018), the same dissociation. So it is not generic spacing or parenthesis sensitivity; specifically the multiplicative composition breaks. (A1/A2 are instruct-only; we do not extrapolate to base.)

It is controlled, not an artifact. Three further controls hold. *Depth* is a distinct stressor: a redundant nesting $((0 + B)) * C =$ drops to 0.038, but nesting degrades *addition* too (nested addition 0.665, base), so we treat depth as a general parsing stressor, not more composition evidence. *Magnitude* is ruled out: within every matched- $|B \cdot C|$ bin C1 stays well below C4 (0.53 vs 0.96 smallest, 0.00 vs 0.70 largest). And the collapse is an *elicitation* fragility: two to four fixed-seed in-context examples recover C1 to the C4 ceiling (instruct 0.265 $\rightarrow$ 0.885 [.85, .91] $\rightarrow$ 0.915 [.88, .94]; base 0.508 $\rightarrow$ 0.838 $\rightarrow$ 0.882), so it is not an inability to multiply.

Two overstatements we avoid. The effect is neither universal (base $(B + 0) * C$ barely moves, drop 0.015, vs 0.33 for $(0 + B)$; Appendix B) nor uniformly worse under tuning (on plain parentheses instruct is *better*, 0.755 vs 0.495); and C0, not 1.0, is the reference ceiling ($B * C =$ is 0.848 instruct).

5 A Dormant Subspace

The seductive story is that the product is computed and represented at the answer position but not read out. We test it in three steps and reject it.

The product decodes (RQ2, correlational). A cross-validated probe recovers $B \cdot C$ from the “=” residual at $R^2 = 0.965$ (instruct), 0.967 (base) and the operand B at 0.97, with B decodable at “)” at *every* layer including the last (instruct 0.98, layer 31); taken alone this looks like a represented product.

But the decode is operand-dominated (RQ2, selectivity). Decodability of $B \cdot C$ need not mean the product is represented, since $B \cdot C$ is itself largely linear in B, C over a narrow band. We compare against a linear- (B, C) baseline (Figure 2a). At [20, 49] the baseline reaches $R^2 = 0.968$, equal to the decode, so the gap is -0.0002 : no headroom, nothing beyond the operands. At [2, 99], where $B \cdot C$ is nonlinear in the operands, a small gap appears: $+0.068$, 95% CI [0.051, 0.086] (base) and $+0.060$, [0.043, 0.079] (instruct), both excluding zero, and still positive at a layer pre-registered from the C4 decode peak rather than the C1 argmax. We flag that [2, 99] is a harder regime where the parts fall below our 0.80 floor (C4 = 0.71 base, 0.73 instruct), so we read it as a stress test, not a comfortably in-scope measurement, and rest the headline on the band-independent causal null below. The decode is operand-dominated (≥ 0.94 of it is reproduced by the linear- (B, C) baseline, ≈ 1.0 at [20, 49]) with at most a small genuine component that tracks behavior (the succeeding C4 surface carries a larger gap than the failing C1).

And the decodable direction is causally inert (RQ3). We steer the “=” residual along the fitted direction to write a counterfactual product P' , on held-out items. The full-product log-probability change brackets zero at every layer, equal to norm-matched random and shuffled controls (Appendix D); this is the discriminative null. (The flip criterion, does the model emit P' , also stays zero

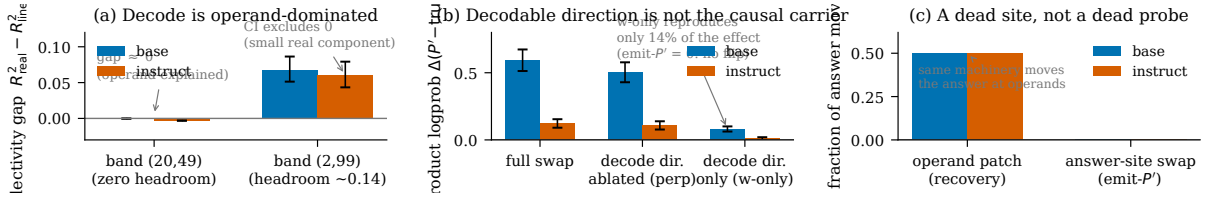


Figure 2: The decodable answer-site subspace is dormant (bars are 95% CIs). **(a)** Selectivity gap (decode R^2 for $B-C$ at “=” minus a linear- (B, C) baseline): at [20, 49] no headroom (gap ≈ 0 , operand-explained); at [2, 99] a small gap whose CI excludes zero. **(b)** Causal share of a full-residual swap at “=” (base): ablating the decode direction (perp) preserves most of the effect, the decode direction alone (w-only) reproduces little, CIs non-overlapping; the swap flips no answers (emit- $P' = 0$). **(c)** The dissociation as a fraction of the answer moved (two metrics): patching moves the answer at the operand positions (recovery 0.50) but the answer-site swap flips nothing, a dead *site*, not a dead probe.

but saturates, staying zero even for a reference intervention that *does* move the log-probability, so we do not lean on it.) A dead direction could be a dead instrument, so we report a positive control, noting it uses a different readout and operation: patching the operand positions under a clean-answer recovery metric moves the output substantially (recovery 0.50, attribution ≈ 2.0), whereas the answer-site interventions do not. Even a full-residual swap at “=” fails to flip the answer (emit- $P' = 0$), at the late layer where it has the most effect it shifts the log-probability by only +0.59 (base).

Certifying the illusion. We decompose that swap effect into the decode direction and its orthogonal complement (Figure 2b). Ablating the decode direction still leaves +0.50 (base), while the decode direction alone contributes only +0.08, with non-overlapping CIs: the small sub-flip variance the swap moves lies almost entirely orthogonal to the decodable direction. (The instruct swap effect is smaller, +0.12, its decode-direction component not significantly positive; we report the decomposition for base.) This locates where the swap’s weak effect lives, not a usable causal handle, since no intervention flips the answer: the dormant subspace signature of Makelov et al. (2024) made causal, the direction a probe finds is not the one the model uses. Consistent with Stolfo et al. (2023) and Nikankin et al. (2025), the operands carry the signal but no sparse head set reconstructs it (best head < 0.02); the value at “=” decodes and correlates yet cannot be steered.

6 Generality and Limitations

Replication in matched format, and a format confound. In matched bare format the collapse is not specific to Llama-3.1-8B. Among models that

clear a parts floor ($C4, C6 \geq 0.80$), the composed form collapses for Gemma-2-9b-it ($C1 = 0.043$, $C4 = 0.99$) and Llama-3.2-3B ($C1 = 0.36$), alongside base and instruct Llama-3.1-8B; three others (Qwen2.5-7B, Mistral-7B, Llama-3.2-1B) fail the parts in bare format and are out of scope, not counterexamples (we name rather than drop them). Chat formatting changes this, not cleanly: Gemma-2-9b-it composes in its native chat template ($C1 = 0.95$), which would look like a family dissociation, but there Llama-3.1-8B falls below the parts floor ($C4 = 0.70$) and leaves scope, so the composing and failing models are never compared in the same format. We read this as replication in matched bare format plus a prompt-format confound, *not* as evidence that some families are immune; per-model native-format evaluation is the right axis for a clean cross-family claim, which we do not make.

Limitations. The causal localization is distributed (operand positions matter but no sparse circuit is recovered), so we characterize *where*, not a minimal mechanism; the operation and depth controls (A1, A2, D1) are instruct-only; and the zero-excluding gap is small and only in the harder band, with the dormancy resting on the causal-share certification. None of these affects the central result.

Conclusion. A parenthesized additive identity selectively breaks multiplicative composition in Llama-3.1-8B, and the tempting reading that the product is represented at the answer site but unused is an interpretability illusion: the decodable answer-site subspace is operand-dominated and causally inert, while the answer is re-derived diffusely from the operands. The portable lesson is that a high decodability R^2 needs a baseline and a causal control before it can support any claim about what a model computes; we release code, data, and the controls

so the null and its positive control can be checked.

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A Full Battery and Confidence Intervals

Table 2 gives every condition with Wilson 95% intervals ($N = 400$). Base A1/A2/D1 are not in our committed data and are omitted rather than estimated.

	surface	base	instruct
C0	$B * C =$	.838 [.80,.87]	.848 [.81,.88]
C1	$(\emptyset + B) * C =$	.508 [.46,.56]	.265 [.22,.31]
C2	$(\emptyset + B * C) =$	n/a	.678 [.63,.72]
C3	$(B) * C =$	n/a	.755 [.71,.80]
C4	$(B * C) =$	.890 [.86,.92]	.908 [.88,.93]
C5	$\emptyset + B * C =$	n/a	.542 [.49,.59]
C6	$(\emptyset + B) =$	1.00 [.99,1.0]	1.00 [.99,1.0]
C7	$(\emptyset+B)*C=$	.018 [.01,.04]	.190 [.16,.23]
C8	$(B*C)=$	.765 [.72,.80]	.785 [.74,.82]
A1	$(\emptyset + B) + C =$	n/a	.995 [.98,1.0]
A2	$\emptyset + (B + C) =$	n/a	.928 [.90,.95]
D1	$((\emptyset + B)) * C =$	n/a	.038 [.02,.06]

Table 2: Full battery, exact-match accuracy with Wilson 95% CIs.

Magnitude control. Within matched $|B \cdot C|$ bins (instruct), C1 vs C4: [440, 813] .53/.96 ($n=79$); [813, 1185] .28/.94 (126); [1185, 1558] .16/.89 (104); [1558, 1930] .19/.89 (64); [1930, 2303] .00/.70 (27). Every paired difference has a positive bootstrap CI.

B Wrapper Blind-Spot Map

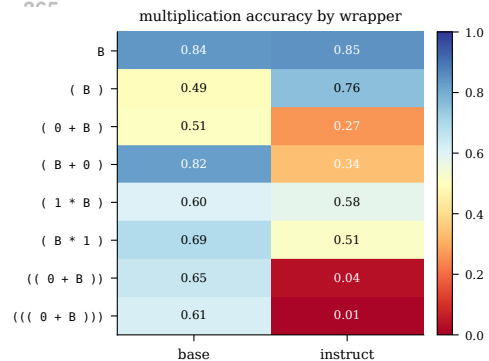


Figure 3: Multiplication accuracy by wrapper. The blind spot is position-dependent (left-placed $(\emptyset + B)$ vs right-placed $(B + \emptyset)$) and not uniformly worse under instruction tuning (plain parentheses).

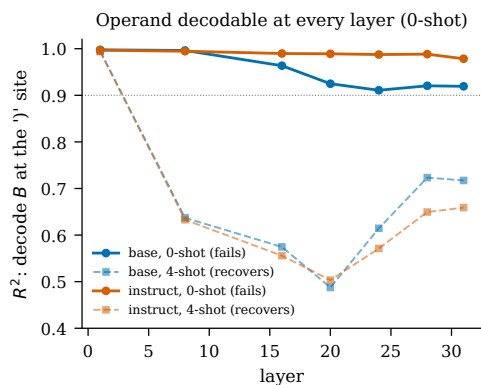


Figure 4: Operand B decodes at every layer in the failing zero-shot regime (solid). The few-shot curves (dashed) drop mid-network; shown for completeness only, and we draw no mechanistic conclusion from the few-shot decodability.

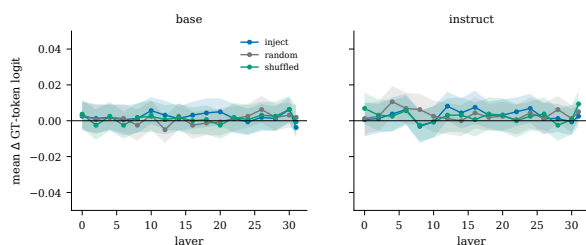


Figure 5: Probe-direction steering at “=” is inert at every layer for inject, random, and shuffled targets (mean Δ ground-truth-token logit, 95% CIs through zero). This first-token metric is a null; the discriminative full-product re-score is Figure 2b.

C By-Layer Decodability 399

D Steering Nulls 400

E Cross-Model Battery 401

Matched bare format, in scope if C4 and C6 $\geq$ 402
0.80: Llama-3.1-8B base C1 .508 / instruct .265; 403
Gemma-2-9b-it .043 (C4 .99); Llama-3.2-3B .358 404
(C4 .88). Out of scope (parts fail): Qwen2.5-7B 405
(C4 .00), Mistral-7B (C4 .62), Llama-3.2-1B (C4 406
.31). Chat format: Gemma-2-9b-it composes (C1 407
.95, C4 .99) but Llama-3.1-8B chat falls out of 408
scope (C4 .70); the composing and failing models 409
are never in the same format. 410